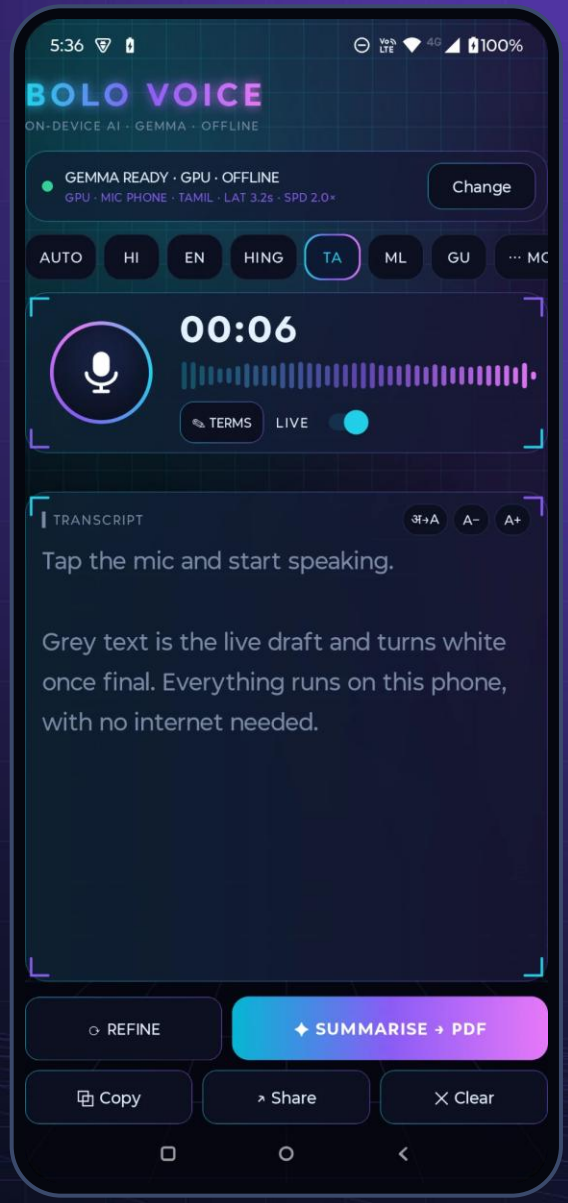
ICAI · AICA LEVEL 2 · PROJECT PRESENTATION

BOLO VOICE

On-device AI speech-to-text for Indian languages: private, offline, built for Chartered Accountants

Records meetings · transcribes live in 21 languages · refines · summarises to PDF, all on the phone with Google's Gemma model

OCTOBER 2026



THE PROBLEM

Why CAs need a different kind of transcription tool

01

Client confidentiality

Cloud transcription uploads client conversations to third-party servers, which sits uneasily with the ICAI Code of Ethics duty of confidentiality and the DPDP Act, 2023.

02

India speaks many languages

Real meetings mix Hindi, English, Hinglish and regional languages. Most tools are tuned for English and translate or garble Indian speech.

03

Field work is often offline

Stock verification, factory and site visits, and hearings often have weak or no network. Online tools simply stop working.

Result: CAs either take notes by hand, losing detail, or accept privacy risk to use cloud AI. Bolo Voice removes that trade-off.

THE SOLUTION

Bolo Voice at a glance

100%

on-device processing

audio never leaves the phone

21

languages supported

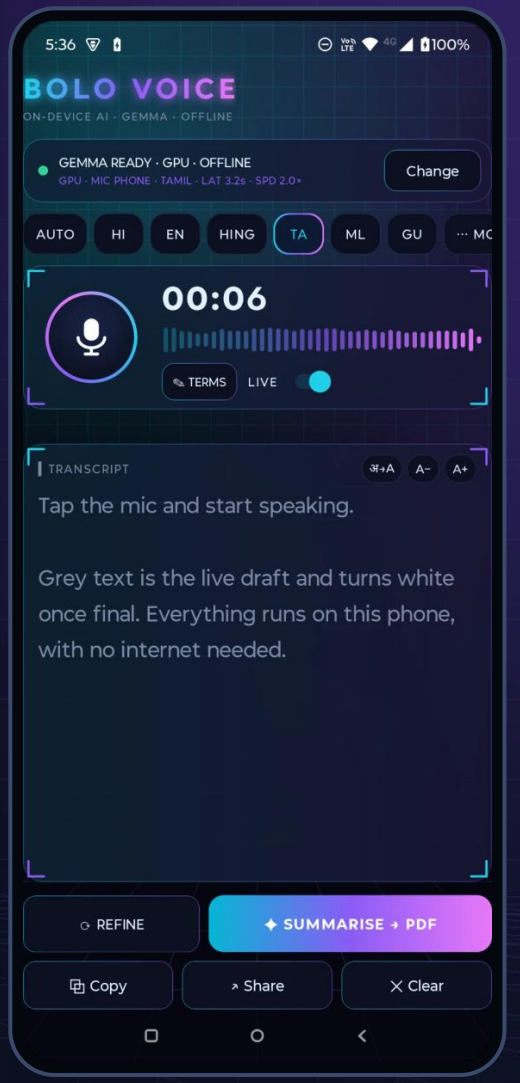
6 tuned: HI · EN · Hinglish · TA · ML · GU

0

internet needed

works in flight mode, no API key, no login

REAL SCREENSHOT · THINKPHONE 25



INTERFACE TOUR

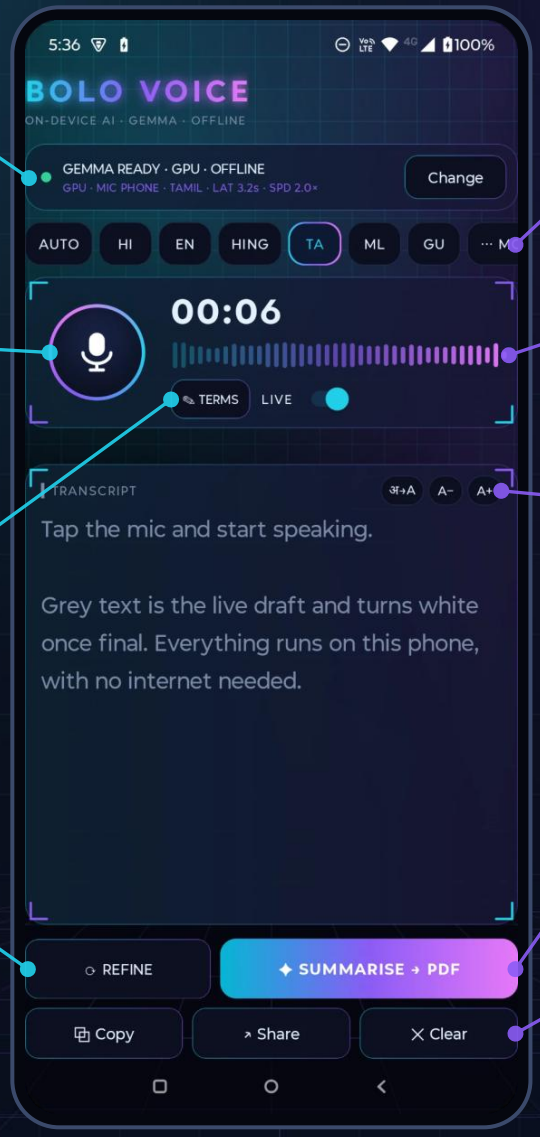
Every part of the screen has one job

1 **Engine status & live stats**
Model ready on GPU; mic, language, latency, speed

3 **Neon record button**
Tap to record; radial bars react to your voice

5 **✎ Terms & LIVE drafts**
Names spelled right; grey text while you speak

7 **⟳ Refine**
Re-transcribe the whole recording for accuracy



2 **One-tap language chips**
AUTO · HI · EN · HING · TA · ML · GU · more

4 **Timer & waveform**
Live mic level and recording time

6 **Transcript tools**
अ→A Roman view · A– / A+ text size

8 **✦ Summarise → PDF**
Four-section summary, save or share

9 **Copy · Share · Clear**
Send the transcript anywhere

KEY FEATURES

Eight features that make it practical for CA work

01

Live transcription

Grey drafts every 3 s; final text at natural pauses

02

21 Indian languages

Six tuned; each written in its own script

03

Smart auto-detect

Locks onto the language after two agreeing segments

04

⟳ Refine pass

Second, higher-accuracy pass over the whole recording

05

Summary → PDF

Overview, key points, decisions, action items

06

Terms glossary

Client names and jargon spelled exactly right

07

Timestamped paragraphs

[02:14] markers at every real pause

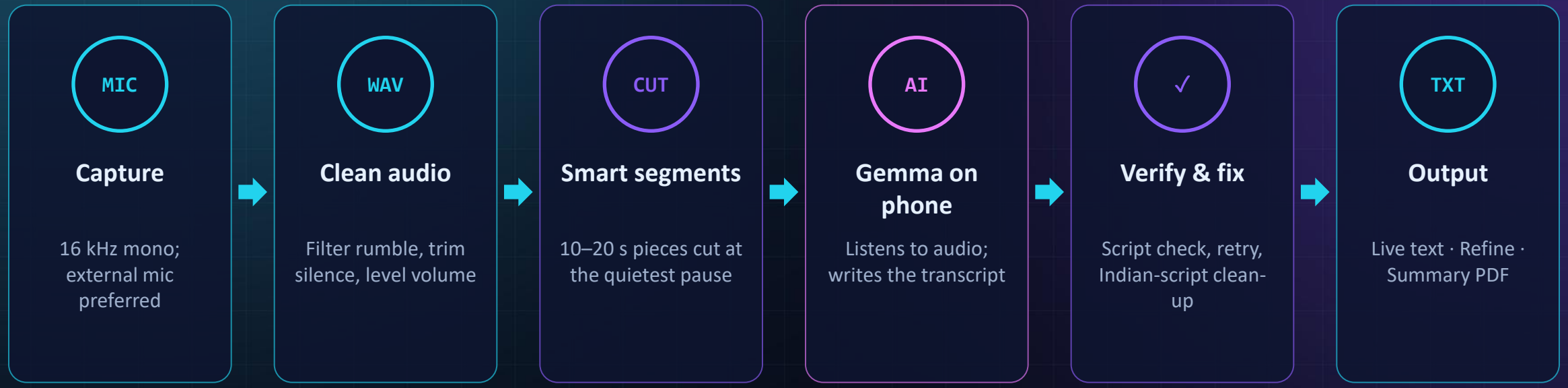
08

Roman view & mic support

अ→A for non-readers; USB-C/wired mics auto-used

HOW IT WORKS

From voice to verified text: the processing pipeline



THE DESIGN PRINCIPLE

Never trust the model blindly. Clean what goes in (audio), check what comes out (script and words), and keep the human in charge of the final version.

Google's Gemma, running entirely on the phone

- **Model:** Gemma 4 E2B, Google's open model (Apache-2.0), about 3.6 GB, loaded once from phone storage
- **Multimodal:** it listens to the audio directly; no separate speech engine
- **Runtime:** Google LiteRT-LM uses the phone's GPU, falling back to CPU automatically
- **Prompt engineering:** per-language instructions with real code-mixed example sentences
- **Deterministic output:** greedy decoding (top-k = 1) for verbatim, repeatable text
- **Warm-up:** a tiny practice run at load makes the first real segment fast

3.2 s — time to transcribe a segment

2.0× — faster than real time (GPU)

4,096 — token context per request

MEASURED ON THINKPHONE 25 · 8 GB RAM

LOGIC, PART 1

Audio intelligence: clean input means accurate output

1

Rumble filter

80 Hz high-pass removes AC, fan and traffic hum

2

Speech detection

Learns the room's noise floor; skips noise-only audio

3

Silence trimming

Ends trimmed; pauses over 1 s shortened to 0.5 s

4

Loudness levelling

Soft or distant voices raised up to 8×

5

Pause-aware cuts

Segments end at the quietest 100 ms, not mid-word

WHY IT MATTERS

Speech AI tends to invent words when it hears silence or noise. Never sending silent audio is the single biggest guard against made-up text.

TESTED

Unit tests confirm: rumble cut by more than 80% with voice kept above 90%, 12 s of audio trimmed to about 3 s, quiet speech boosted more than 4× without distortion.

Language intelligence built for India

AUTO | HI | EN | HING | TA | ML | GU | ⋯

SAME SENTENCE, OWN SCRIPT, ENGLISH WORDS KEPT

HI	कल की meeting में हम budget discuss करेंगे
TA	நாளைக்கு meeting-ல budget பத்தி பேசலாம்
ML	നാളെ meeting-ൽ budget ചർച്ച ചെയ്യാം
GU	કાલે meeting માં budget ની ચર્ચા કરીશું
HING	Kal ki meeting mein hum budget discuss karenge

- **Tuned prompts:** each language gets an example in its own script, plus rules (e.g. Gujarati, never Devanagari)
- **Script check:** if the output is in the wrong script or translated, it's retried with a firmer instruction
- **Auto-detect lock:** the language is fixed once two segments agree, so no mid-meeting switching
- **Hinglish vs English:** told apart by common Hindi words (hai, nahi, kya, abhi…)
- **Script clean-up:** correct । full stop, Malayalam chillu letters, Unicode normalisation

RESPONSIBLE AI

Guardrails: every AI risk has a control

AI RISK	CONTROL BUILT INTO THE APP
Hallucination on silence	Silent or noise-only audio is never sent to the model
Translation instead of transcription	Script check rejects wrong-script output and retries
Runaway repetition loops	Repeated phrases collapsed; output capped at 512 tokens
AI rewording the client's words	Polish accepted only if at least 97% of words match exactly
Over-reliance on automation	Grey drafts are marked; Refine asks Replace or Keep

PRIVACY & COMPLIANCE

Your client's words never leave the phone



bytes sent to any server

The app has no internet permission at all. Recording, transcription, summary and PDF are all created and stored on the device.

- **No cloud processor**
  no third-party access to client conversations
- **ICAI Code of Ethics**
  supports the duty of confidentiality of client information
- **DPDP Act, 2023**
  data minimisation, with personal data processed only on the device
- **User control**
  one tap clears the transcript and the stored recording
- **Audit trail**
  timestamped transcripts support documentation of discussions

USE CASES

Where it fits in CA practice

01

Client meetings & minutes

Record, refine and share a four-section summary PDF the same day

02

Audit walkthroughs & inquiries

Capture management's explanations verbatim for working papers

03

Tax & GST hearings

Timestamped notes of submissions and officers' observations

04

Board & committee meetings

First draft of minutes with decisions and action items

05

Due diligence & internal audit

Confidential interviews stay on the device

06

Dictation in your own language

Dictate file notes in Hindi, Gujarati, Tamil or Hinglish

SCENARIOS

Real situations it is built for

SCENARIO	LANGUAGE	WHY BOLO VOICE FITS
Stock verification at a factory	**Hindi / Hinglish**	No network on the shop floor; works fully offline
GST client meeting in Surat	**Gujarati**	Tuned Gujarati, never written in Devanagari
Audit closing meeting in Chennai	**Tamil + English**	English terms kept; Tamil script preserved
Family business review in Kochi	**Malayalam**	Correct chillu letters; Roman view for colleagues
Confidential M&A due diligence	**English**	Nothing uploaded; one tap clears everything

WORKFLOW

From conversation to shareable PDF in five taps

1

Pick language

Tap HI, TA, GU… or AUTO

2

Record

Live grey drafts as people speak

3

Stop & Refine

Higher-accuracy second pass

4

Summarise

Choose English or the spoken language

5

Save / Share PDF

Email, WhatsApp, Drive

OUTPUT PDF CONTAINS

- Overview, key points, decisions
- Action items with owners
- Date and recorded duration
- Full timestamped transcript

HONEST ASSESSMENT

Current limits and the roadmap

CURRENT LIMITS

- **Hardware:** needs a phone with 8 GB RAM; the model is a 3.6 GB one-time download
- **Speakers:** does not yet label who said what
- **Rarer languages:** Konkani, Sindhi, Maithili are weaker than the six tuned ones
- **iPhone:** app builds, but installing needs Apple signing
- **Review:** output is a draft; a professional must review it

ROADMAP

- **Speaker labels:** identify each participant in the transcript
- **Accuracy benchmark:** measure word error rate per language on real CA meetings
- **Export:** minutes straight to Word or Excel templates
- **iPhone release:** through Apple TestFlight
- **Domain terms:** built-in glossary of accounting, GST and audit terms

Principle: Bolo Voice drafts, and the Chartered Accountant decides. AI output is never the final record.

UNDER THE HOOD

Technology stack

01

AI MODEL

Gemma 4 E2B (Google, Apache-2.0), on-device, 4-bit

02

AI RUNTIME

Google LiteRT-LM 0.18.1: GPU (OpenCL) with CPU fallback

03

APP

Native Android: Kotlin 2.4, Android 12+, about 55 MB APK

04

AUDIO

16 kHz PCM, Butterworth filter, adaptive speech detection

05

LANGUAGES

21 languages, 11 scripts, ICU transliteration for Roman view

06

QUALITY

38 automated unit tests: audio, scripts, polish safety, timestamps

KEY TAKEAWAYS

1

Private by design

AI on the device: client conversations never leave the phone

2

Built for India

Six tuned languages, own scripts, code-mixing handled

3

Trustworthy AI

Input cleaning, output verification, human in the loop

Thank you · Questions?

